

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
4	(a)		Diagram showing relevant right-angled triangle with d and θ labelled (1) $d \sin \theta$ stated to be path difference [between light from adjacent slits] (1) $d \sin \theta = n\lambda$ is condition for constructive interference [or for light to 'arrive' in phase (1) mention of constructive interference only don't award the mark	3			3	1	
	(b)	(i)	Scales correctly chosen (more than $\frac{1}{2}$ used) and orientated and axes clearly labelled (1) $\sin \theta$ plotted correctly, whether or not tabulated - allow $\pm \frac{1}{2}$ small square tolerance (1) Best straight line through origin or by eye (1) If θ plotted: don't award the scales mark, plotting of points correctly mark can be awarded and best fit curve or straight line can be awarded		3		3	3	3
		(ii)	Gradient = $\frac{\lambda}{d}$ or single point on graph taken by implication (1) Gradient = 0.285 [$\pm$ 0.005] (1) $\lambda = 510$ [$\pm$ 15] nm (1) unit mark If θ plotted: If λ determined correctly award a maximum of 2 marks			3	3	3	3
			Question 4 total	3	3	3	9	7	6